Pricing Analysis Case Study

In this case, we see that there is a rideshare organization (hereby referred to as simply RS) that is attempting to optimize their pricing in order to maximum their revenue in a 1-year timespan. Additionally, they must contend with the fact that if their drivers receive less pay, customers are less likely to match with a driver. Failure to match with a driver increases the likelihood that a customer will churn and refuse to utilize the service anymore. A previous pricing experiment has been performed which demonstrates the aforementioned fact.

There is an indication that there is a prevailing wage and prevailing rate that cannot be modified, as indicated by the question: **"how much more or less do you pay drivers per trip (by changing RS's take)?"** Thus, we maintain those pricing rates and associated churn rates and make no modifications to them. In a realistic scenario, paying the drivers less than the prevailing wage would increase driver churn and lower driver rides/month, which would further decrease revenues due to reducing match rates further than what is provided in the case (subsequently increasing rider churn rates). Meanwhile, increasing the ride rate to greater than the prevailing rate would likely decrease demand (and the extent of the pricing experiment does not allow us to properly analyze how dramatically this demand would decrease).

The case does not explicitly state targeted acquisition rates, so we therefore use the following table as an approximation:

Low Acquisition Rate		Medium Acquisition Rate		High Acquisition Rate	
Rider CAC	$10	Rider CAC	$15	Rider CAC	$20

These allow us to determine a rough estimate of what revenue we may anticipate based on the relevant acquisition rates.

In the case, we are provided with the following table of information:

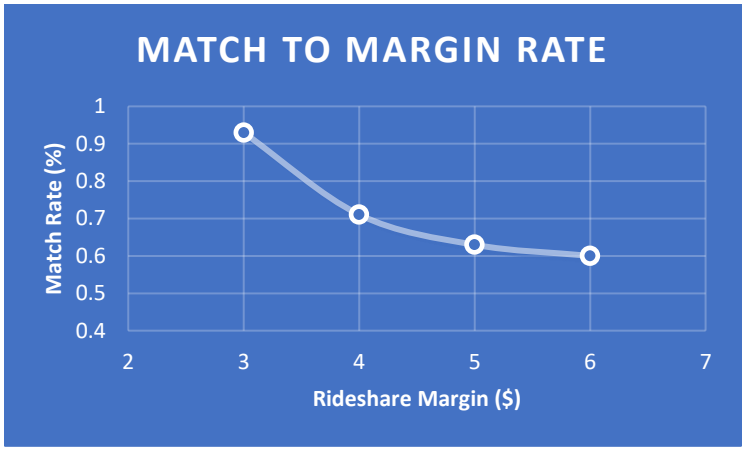
Item	Data
Driver Acquisition	-$500
Driver Churn	5%
Driver rides/month	100
Rider Acquisition	$10-$20
Rider rides/month	1
Rider churn w/o issue/month	10%
Rider churn w/ issue/month	33%

Of note, the rate at which riders find a matching ride appears to be variable depending on the margin that RS receives. However, it is not a direct linear scale and instead appears to be an exponential decay function. For the purposes of this analysis, I have treated this exponential decay as:

$$Match\ Rate = a + be^{-RS\ Margin}$$

Afterwards, utilizing Excel's solver function for match rates of 0.6 and 0.93 with respective Rideshare margins of 6 and 3, we find the following function:

$$Match\ Rate = 0.582709 + 6.975518e^{-RS\ Margin}$$



The above chart provides a visual representation of the curve at various margins for RS and the associated match rates.

This problem fundamentally follows a lifetime value of a customer model, where the lifetime value of a customer is defined as:

$$Customer\ Lifetime\ Value = \sum_{t=0}^{N} \frac{(p-c)r^t}{(1+i)^t}$$

where N = number of years, p=revenue customer generates in year t, c = variable cost of a customer in year t, r = retention rate, i = interest rate. Since the margin is relatively fixed over the course of the one period and we are dealing with just one year, we can simplify this function by assuming an infinite economic life and convert it to:

$$Customer\ Lifetime\ Value = m\frac{r}{1-r+i}$$

Where m is the margin (created by taking p – c of a customer). For the purposes of this analysis, we shall treat the interest rate as 10% as a quick rule of thumb.

As we possess situations where riders have two churn rates, it is more relevant to calculate the churn rates of those two customer bases separately. Additionally, since these are very high churn rates, it is better to utilize the following equation to measure annual churn:

$$Annual\ Churn = 1 - (1 - Monthly\ Churn)^{12}$$

From this, we find the annual churn rates for customers who experience no matching issues as 72%. Meanwhile, the annual churn rate for customers who do experience matching issues as 99%. In both cases, it would be recommended that RS re-examine their churn measurements to not be focused on customer churn rates. The justification for this is that consumers do not regularly utilize

services such as RS on a monthly basis but are likely to return to it on an annual basis. It would be much more relevant to a firm such as Rideshare to instead measure based on monthly recurring revenue churn, as that provides a much better understanding of the financial health of the firm over a specific period as opposed to customer churn rates. However, since that data is lacking, we shall continue with what is available by calculating the retention rate as:

$$Annual\ Retention\ Rate\ (r) = 1 - Annual\ Churn$$

Utilizing all the information as provided and entered into an excel spreadsheet, we are able to subsequently calculate the CLV and additionally calculate the CLV/CAC ratios for low, medium, and high acquisition rates. We also calculate a weighted ratio using the match rates as the weights as follows:

$$Weighted\ Ratio = \frac{\sum weights * CLV/CAC}{\sum weights} \text{ or } Weighted\ Ratio = \frac{\sum weights * CLC}{\sum weights}$$

Combining all the information, we generate this final table:

RS Margin	Rider Type	CLV	Low CLV/CAC	Med CLV/CAC	High CLV/CAC
$6					
	w/o error	$24.87	2.49	1.66	1.24
	w/ error	$0.54	0.05	0.04	0.03
Weighted Ratio (60%/40%)		$15.14	1.51	1.01	0.76
$5.5					
	w/o error	$22.80	2.28	1.52	1.14
	w/ error	$0.49	0.05	0.03	0.02
Weighted Ratio (61.12%/38.88%)		$14.13	1.41	0.94	0.71
$5					
	w/o error	$20.73	2.07	1.38	1.04
	w/ error	$0.45	0.04	0.03	0.02
Weighted Ratio (62.97%/37.03%)		$13.22	1.32	0.88	0.66
$4.5					
	w/o error	$18.65	1.87	1.24	0.93
	w/ error	$0.40	0.04	0.03	0.02
Weighted Ratio (66.02%/33.98%)		$12.45	1.25	0.83	0.62
$4					
	w/o error	$16.58	1.66	1.11	0.83
	w/ error	$0.36	0.04	0.02	0.02

Weighted Ratio (71.05%/28.95%)		$ 11.89	1.19	0.79	0.59
$3.5					
	w/o error	$14.51	1.45	0.97	0.73
	w/ error	$0.31	0.03	0.02	0.02
Weighted Ratio (79.33%/20.67%)		$11.58	1.16	0.77	0.58
$3					
	w/o error	$12.44	1.24	0.83	0.62
	w/ error	$0.27	0.03	0.02	0.01
Weighted Ratio (93%/7%)		$11.58	1.16	0.77	0.58

In every situation, one can see that that is a decreasing ratio of CLV/CAC for riders regardless of the reduced churn rates associated with decreasing margins for RS. This trend also holds when utilizing a linear match to margin rate ratio (this data not included in this analysis to sake of brevity). The lifetime values associated with every driver are not particularly relevant to this analysis, as they will maintain consistent margins regardless of acquisition rates, will always remain a net positive, and will follow similar downwards trends of CLV/CAC ratios as the riders if their churn rates do not decrease with decreased RS margins. At any point where the CLV/CAC ratio is lower than 1.0, it indicates that there will be a profit loss for RS as acquisition costs will be more than the lifetime value expected of each rider. This will be buoyed by the drivers of the service generating positive profits but will make the overall rider segment unprofitable. RS should prioritize the option where the highest number of CLV/CAC ratios are greater than 1.0, prioritizing the examination of the weighted ratios.

Therefore, the best option for Rideshare in the 1-year time period is to **maintain their margin at $6**. However, this suggestion comes with the following caveats for executives at RS:

- In the short term, this will increase revenues for RS. However, it is anticipated to cause reputational harm to RS and may decrease demand for the service in the long-term.
- The present churn rates at RS do not adequately capture customer usage as people do not continue to purchase ride-shares monthly. It is recommended to capture data on monthly recurring revenue churn to better capture overall health of the service.
    - Customer churn rates are also likely to be inaccurate and more likely to be in a form such as churn(riders) = function(match), where the first few failures have a more pronounced impact on churn rates than later failures.
- There should be an experiment performed to determine what influences driver churn rates, as there needs to be an appropriate number of drivers to provide service to riders. Can the drivers service even more riders if given incentives at different ride breakpoints?

- Seasonality effects should be accounted for in future analyses, as rideshare services will spike at different periods (i.e., college spring breaks, summer vacations, Christmas gatherings, etc.)